Huawei Access Node (JAY-ACC-35) Routing Protocol Flap Troubleshooting SOP

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5000

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Huawei Access Node, JAY-ACC-35, located at the Jayanagar Metro Site. In Huawei's eLTE (Enhanced Long-Term Evolution) architecture, the Access Node plays a pivotal role in ensuring seamless data transmission between the wireless and wired networks. The Routing Protocol Flap failure domain is primarily related to the instability in the routing information exchanged between the Access Node and its neighboring nodes. This instability can be triggered by various factors, such as BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. In this SOP, we will navigate through the Huawei CLI to diagnose and resolve the root causes of this failure domain.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site to facilitate physical access to the JAY-ACC-35 node.

- Verify the status of the redundant links connected to the node and ensure they are operational.

- Check the environmental conditions at the site, including temperature, humidity, and power supply, to rule out any external factors contributing to the failure.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table all, Expected Output: BGP routing table information, including the current routing table and the number of routes in each table, Description: This command displays the BGP routing table and helps identify any BGP route churn issues.

Command: display ospf neighbor, Expected Output: Information about OSPF neighbors, including their state and the number of neighbors, Description: This command displays OSPF neighbor information and helps diagnose any adjacency loss issues.

Command: display isis database, Expected Output: ISIS database information, including the current LSP count and sequence numbers,
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```
Description: This command displays the ISIS database and helps
identify any LSP flooding issues.
```

Resolution Steps

Step 1: Step: Clear BGP route churn, Command: clear bgp all, Description: This command clears all BGP routes and helps resolve BGP route churn issues.

Step 2: Step: Verify OSPF adjacency, Command: ver OSPF neighbor, Description: This command verifies OSPF adjacency and helps diagnose any adjacency loss issues.

Step 3: Step: Check ISIS LSP sequence numbers, Command: display isis database sequence, Description: This command checks ISIS LSP sequence numbers and helps identify any LSP flooding issues.

Post-Check Verification

```
Command: display bgp routing-table all, Expected Output: Verified BGP
routing table with no issues, Description: This command verifies the
BGP routing table and ensures there are no issues.

Command: display ospf neighbor, Expected Output: Verified OSPF
adjacency with no issues, Description: This command verifies OSPF
adjacency and ensures there are no issues.

Command: display isis database, Expected Output: Verified ISIS
database with no issues, Description: This command verifies the ISIS
database and ensures there are no issues.
```

Roll-back Procedures

- Step: Restore BGP routing table, Command: reset bgp all, Description: This command restores the BGP routing table to its previous state and helps roll back any changes made during the troubleshooting process.

- Step: Restore OSPF adjacency, Command: reset OSPF neighbor, Description: This command restores OSPF adjacency to its previous state and helps roll back any changes made during the troubleshooting process.

- Step: Restore ISIS LSP sequence numbers, Command: reset isis database sequence, Description: This command restores ISIS LSP sequence numbers to their previous state and helps roll back any changes made during the troubleshooting process.